

SUPPLEMENTARY MATERIAL

Table S1. Sampling points, treatment, coordinates, sampling year, impact index, and distance (m) from the nearest pit to the streams sampled in the Carajás National Forest. Sampling 1 (September/October 2015). Sampling 2 (October/November 2016), Sampling 3 (October 2017).

Sampling point (stream)	Treatment	Coordinates		Year	Impact Index	Distance from the nearest mining pit (m)	Average area of nearest mining pit (m ²)*
		X	Y				
CONT-01AG	Preserved	589329.9	9318463.1	2015	0.501	8399.5	22,166,106
CONT-02AG	Preserved	592535	9316425.5	2015	0.345	9653.1	22,166,106
CONT-03AG	Preserved	596393.6	9315166.1	2015	0.267	10186.8	22,166,106
CONT-04AG	Preserved	591046.9	9323168.5	2015	1.926	3914.9	22,166,106
CS11-01AG	Preserved	565199.5	9300429.1	2015	0.156	10549.8	13,388,681
CS11-02AG	Preserved	562222.4	9299915.4	2015	0.118	12448.6	13,388,681
CS11-03AG	Preserved	563492.4	9297517.6	2015	0.161	10092.4	13,388,681
CS11-05AG	Preserved	561918.6	9296287.3	2015	0.136	11034.2	13,388,681
MANG-01AG	Altered	574393.7	9325717.3	2015	37.851	471.3	8,376,965
MANG-02AG	Altered	576608.2	9322851.8	2015	44.584	434.3	8,376,965
MANG-03AG	Altered	579584.8	9324979.1	2015	152.184	234.8	8,376,965
MANG-04AG	Altered	578537.1	9328622.4	2015	84.466	315.4	8,376,965
N4-01AG	Altered	588245.5	9325868.8	2015	7.688	1751.1	22,166,106
N4-02AG	Altered	592794.4	9325607.8	2015	10.898	1654.8	22,166,106
N4-03AG-R	Altered	592125.1	9324559.8	2015	4.092	2697	22,166,106
N4-04AG	Altered	588546.3	9328361.8	2015	16.558	1175.2	22,166,106
N5-01AG	Altered	595263.1	9325711.9	2015	31.584	563	9,156,887
N5-02AG	Altered	595874.3	9324910.2	2015	32.508	543.7	9,156,887
N5-03AG	Altered	598104.8	9326712	2015	14.866	821.2	9,156,887
N5-04AG	Altered	598200	9327196.2	2015	9.623	1051.2	9,156,887
S11D-01AG	Altered	575669	9294108.2	2015	16.958	889.4	13,388,681
S11D-02AG	Altered	573089.5	9295355.1	2015	2.376	2391.2	13,388,681
S11D-04AG	Preserved	569783.	9294713.	201	1.277	3277.6	13,388,681

S11D-05AG	Preserved	566035.4	9290842.1	2015	0.355	6351.4	13,388,681
CONT-01AG	Preserved	589329.9	9318463.1	2016	0.796	6662.4	26,124,097
CONT-02AG	Preserved	592535	9316425.5	2016	0.56	8176.1	26,124,097
CONT-03AG	Preserved	596393.6	9315166.1	2016	0.386	10022.5	26,124,097
CONT-04AG	Altered	591046.9	9323168.5	2016	10.019	1664.8	26,124,097
CS11-01AG	Preserved	565199.5	9300429.1	2016	0.163	10549.8	13,569,335
CS11-02AG	Preserved	562222.4	9299915.4	2016	0.124	12448.6	13,569,335
CS11-03AG	Preserved	563492.4	9297517.6	2016	0.167	10092.4	13,569,335
CS11-05AG	Preserved	561918.6	9296287.3	2016	0.142	11034.2	13,569,335
MANG-01AG	Altered	574393.7	9325717.3	2016	38.702	471.3	8,560,023
MANG-02AG	Altered	576608.2	9322851.8	2016	45.591	434.3	8,560,023
MANG-03AG	Altered	579584.8	9324979.1	2016	155.573	234.8	8,560,023
MANG-04AG	Altered	578537.1	9328622.4	2016	86.347	315.4	8,560,023
N4-01AG	Altered	588245.5	9325868.8	2016	18.65	1200.7	26,124,097
N4-02AG	Altered	592794.4	9325607.8	2016	47.772	778.7	26,124,097
N4-03AG-R	Altered	592125.1	9324559.8	2016	5226.311	70.7	26,124,097
N4-04AG	Altered	588546.3	9328361.8	2016	40.605	807.6	26,124,097
N5-01AG	Altered	595263.1	9325711.9	2016	43.312	502.9	10,122,325
N5-02AG	Altered	595874.3	9324910.2	2016	54.948	437.1	10,122,325
N5-03AG	Altered	598104.8	9326712	2016	16.943	818.7	10,122,325
N5-04AG	Altered	598200	9327196.2	2016	11.157	1048.8	10,122,325
S11D-01AG	Altered	575669	9294108.2	2016	17.192	889.4	13,569,335
S11D-02AG	Altered	573089.5	9295355.1	2016	2.413	2391.2	13,569,335
S11D-04AG	Preserved	569783.3	9294713.7	2016	1.299	3277.6	13,569,335
S11D-05AG	Preserved	566035.4	9290842.1	2016	0.364	6351.4	13,569,335
CONT-01AG	Preserved	589329.9	9318463.1	2017	0.842	6501.4	26,619,508
CONT-02AG	Preserved	592535	9316425.5	2017	0.571	8176.1	26,619,508
CONT-03AG	Preserved	596393.6	9315166.1	2017	0.393	10022.5	26,619,508
CONT-04AG	Altered	591046.9	9323168.5	2017	10.205	1664.8	26,619,508
CS11-01AG	Preserved	565199.5	9300429.1	2017	0.172	10549.7	14,437,368

CS11-02AG	Preserved	562222.4	9299915.4	2017	0.13	12448.2	14,437,368
CS11-03AG	Preserved	563492.4	9297517.6	2017	0.176	10091.8	14,437,368
CS11-05AG	Preserved	561918.6	9296287.3	2017	0.15	11033.5	14,437,368
MANG-01AG	Altered	574393.7	9325717.3	2017	40.56	469.7	8,912,352
MANG-02AG	Altered	576608.2	9322851.8	2017	47.464	434.3	8,912,352
MANG-03AG	Altered	579584.8	9324979.1	2017	161.969	234.8	8,912,352
MANG-04AG	Altered	578537.1	9328622.4	2017	89.895	315.4	8,912,352
N4-01AG	Altered	588245.5	9325868.8	2017	20.889	1143.7	26,619,508
N4-02AG	Altered	592794.4	9325607.8	2017	48.616	778.7	26,619,508
N4-03AG-R	Altered	592125.1	9324559.8	2017	5325.405	70.7	26,619,508
N4-04AG	Altered	588546.3	9328361.8	2017	46.973	757.4	26,619,508
N5-01AG	Altered	595263.1	9325711.9	2017	59.439	430.9	10,175,423
N5-02AG	Altered	595874.3	9324910.2	2017	55.719	437.1	10,175,423
N5-03AG	Altered	598104.8	9326712	2017	17.059	818.7	10,175,423
N5-04AG	Altered	598200	9327196.2	2017	11.243	1048.8	10,175,423
S11D-01AG	Altered	575669	9294108.2	2017	18.291	889.4	14,437,368
S11D-02AG	Altered	573089.5	9295355.1	2017	2.566	2391.2	14,437,368
S11D-04AG	Preserved	569783.3	9294713.7	2017	1.381	3277.3	14,437,368
S11D-05AG	Preserved	566035.4	9290842.1	2017	0.386	6350.6	14,437,368

*Area calculated for the reference month of collection in each year.

Table S2. Functional traits used to build the functional matrix of the Ephemeroptera genera.

Type	Functional trait	Quantitative	Qualitative	Binary	Category	Function	References
Ecology	Functional Feeding Group (FFG)	x			Collector Filterer Scraper Predator Cutter	Balance in the food chain	Merritt and Cummins, 1996, Cummins et al., 2005, Shimano et al., 2012, Cruz, Salles and Hamada, 2013, Brasil et al., 2014, Dedieu et al., 2015.
	Presence of opercular gill		x		1 0	Habitat Resilience, Breathing, Protection	Domínguez et al., 2006, Zhou & Zheng., 2010. Salles et al., 2018.
Morphology	Head size	x			Length	Feeding	Merritt and Cummins, 1996, Goncalves et al, 2003, Shimano et al., 201, Domínguez et al., 2006, Salles et al., 2018.
		x			Width		
	Size of the mesonotum	x			Width	Life cycle (development of the alar weave); nymph size	Domínguez et al., 2006, Flowers & De La Rosa, 2010, Salles et al., 2018.

Table S3. Table of morphological (quantitative) and ecological (qualitative) traits of Ephemeroptera genera used in the data analysis; opercular gill (og), nog: non-opercular gill size.

Traits					
Genus	Size of the mesonotum (mm)	Length head size (mm)	Width head size (mm)	Presence of opercular gill	Feeding group

<i>Amanahyphes</i>	10.927	10.639	12.916	og	Collector
<i>Americabaetis</i>	17.641	16.960	13.551	nog	Collector
<i>Askola</i>	15.268	17.863	17.789	nog	Scraper
<i>Baetodes</i>	18.086	17.966	16.062	nog	Scraper
<i>Brasilocaenis</i>	10.979	0.718	10.089	og	Scraper
<i>Caenis</i>	15.424	0.6335	11.103	og	Scraper
<i>Callibaetis</i>	26.752	24.865	16.135	nog	Collector
<i>Callibaetoides</i>	13.321	0.9958	12.417	nog	Collector
<i>Campylocia</i>	50.676	35.543	39.238	nog	Collector
<i>Cloeodes</i>	15.588	17.956	13.045	nog	Scraper
<i>Farrodes</i>	12.613	16.843	17.962	nog	Scraper
<i>Fittkaulus</i>	12.350	18.333	17.538	nog	Filterer
<i>Hagenulopsis</i>	11.781	15.836	15.489	nog	Scraper
<i>Hydrosmilodon</i>	28.275	34.915	39.132	nog	Filterer
<i>Leptohyphes</i>	15.827	12.812	17.736	og	Collector
<i>Miroculis</i>	14.309	18.704	20.258	nog	Scraper
<i>Paracloeodes</i>	24.072	12.634	18.363	nog	Collector
<i>Paramaka</i>	25.768	29.620	33.622	nog	Filterer
<i>Simothraulopsis</i>	22.461	27.761	24.526	nog	Scraper
<i>Terpides</i>	15.304	13.502	16.671	nog	Filterer
<i>Thraulodes</i>	22.046	20.062	24.583	nog	Scraper
<i>Traverhyphes</i>	23.971	11.882	17.749	og	Collector
<i>Tricorythodes</i>	16.765	0.9332	13.513	og	Collector
<i>Tricorythopsis</i>	22.291	11.860	18.027	og	Collector
<i>Ulmeritoides</i>	13.773	13.085	16.575	nog	Filterer
<i>Waltzoyphius</i>	15.969	14.910	11.333	nog	Collector
<i>Zelus</i>	13.818	0.8606	10.305	nog	Collector

Table S4. List of genera used in functional beta diversity analysis.

Genus	Abundance/Streams		
	Altere d	Preserve d	Tot al
<i>Amanahyphes</i> Salles & Molineri, 2006	14	3	17
<i>Americabaetis</i> Kluge, 1992	29	14	43
<i>Askola</i> Peters, 1969	13	18	31
<i>Baetodes</i> Needham & Murphy, 1924	19	18	37
<i>Brasilocaenis</i> Malzacher, 1986	12	7	19
<i>Caenis</i> Malzacher, 1986	59	12	71
<i>Callibaetis</i> Eaton, 1881	13	1	14
<i>Callibaetoides</i> Cruz, Salles & Hamada, 2013	6	2	8
<i>Campylocia</i> Needham & Murphy, 1924	16	10	26
<i>Cloeodes</i> Traver, 1938	92	59	151
<i>Farrodes</i> Peters, 1971	2	0	2
<i>Fittkaulus</i> Savage & Peters, 1978	217	130	347
<i>Hagenulopsis</i> Ulmer, 1920	1	4	5
<i>Hydrosmilodon</i> Flowers & Domínguez, 1992	95	49	144
<i>Leptohyphes</i> Ulmer, 1919	21	3	24
<i>Miroculis</i> Edmunds, 1963	396	243	639
<i>Paracloeodes</i> Day, 1955	12	3	15
<i>Paramaka</i> Savage & Domínguez, 1992	44	48	92
<i>Simothraulopsis</i> Demoulin, 1966	16	4	20
<i>Terpides</i> Demoulin, 1966	23	35	58
<i>Thraulodes</i> Ulmer, 1920	4	8	12
<i>Traverhyphes</i> Molineri, 2001	37	9	46
<i>Tricorythodes</i> Ulmer, 1920	7	22	29
<i>Tricorythopsis</i> Traver, 1958	10	1	11
<i>Ulmeritoides</i> Traver, 1959	33	54	87
<i>Waltzoyphius</i> Lugo-Ortiz & Mccafferty, 1995	5	2	7
<i>Zelus</i> Lugo-Ortiz & Mccafferty, 1998	134	64	198
Overall Total	1330	823	2153

Table S5- Principal Components Analysis (PCA) results indicating the variables and scores relationships, variance explanation (%) and eigenvalues for principal components 1 and 2.

Variable	PC1	PC2
		-0.89
Conductivity	0.340	8
COD	-0.12	-0.05

	3	2
ItonTot	0.650	0.215
		-0.23
PhosphoTot	0.034	1
		-0.07
MangTot	0.461	8
	-0.31	-0.04
Nitrate	4	1
	-0.02	-0.07
OxygenDis	8	7
pH	0.079	0.248
		-0.88
SolidTot	0.358	6
Temperatur		
e	0.683	0.052
Turbidity	0.725	0.410
Impact.Inde		
x	0.636	0.242
%Explanatio	19.97	16.65
n	7	3
Eigenvalues	2.359	1.967

Table S6. Subset of predictor variables (environmental and spatial dataset) selected by forward selection for explaining the community structure of mayflies in sampled streams.

Matrix	Variable	Accum. R ² _{cum}	Accum. R ² _{adj}	F	p
Environment	Impact Index	0.0351	0.0193	2.222	0.000
				2	3
	Temperature Dissolved oxygen	0.0730	0.0421	2.451	0.000
				2	1
Space	MEM2	0.0467	0.0310	1.606	0.025
				9	3
	MEM9	0.0838	0.0532	2.986	0.000
				1	0
	MEM7	0.1194	0.0747	2.428	0.000
				4	1
	MEM1	0.1533	0.0949	2.391	0.000
				5	1
	MEM4	0.1812	0.1093	2.321	0.000
				3	1
	MEM3	0.2056	0.1204	1.937	0.002
				6	8
				1.719	0.012
				9	4

Table S7. Partial distance-based RDA loadings for models in which environmental and spatial variables selected by forward selection are significative.

Partition	Variable	F β_{sor}		F β_{ness}	
		dbRDA	dbRDA	dbRDA	dbRDA
		1	2	1	2
Environment	Dissolved oxygen	-0.056	-0.005	-0.056	0.295
	Temperature	0.408	0.126	0.430	-0.041
	Impact Index	-0.145	0.271	-0.177	-0.007
Space	MEM1	-0.253	0.174	-	-
	MEM2	0.038	0.024	-	-
	MEM3	-0.226	-0.252	-	-
	MEM4	-0.094	0.218	-	-
	MEM7	0.244	0.281	-	-
	MEM9	0.337	-0.160	-	-

Table S8. Results of Fourth-corner analysis showing the relationships between environmental variables and traits for Ephemeroptera genera. Environmental variables: electrical conductivity (Cond), chemical oxygen demand (COD), total iron (Irontot), total phosphorus (Phosphotot), total manganese (Mangtot), nitrate, dissolved oxygen

(Oxygendis), temperature (Temp), total solids (SolidTot), pH and turbidity. Functional characteristics: tc1 (head length size), tc2 (head width), tm (mesonotum size), gt= trophic guild, scraper, collector, filterer, opercular gill (og), non-opercular gill (nog).

Ite m	Environmental variables	Traits	Observe d	Std. deviation observed	p
1	Cond	tm	0.022	0.793	0.429
2	COD	tm	0.001	0.105	0.905
3	IronTot	tm	-0.013	-0.358	0.729
4	PhosphoTot	tm	-0.020	-0.606	0.546
5	Mangtot	tm	-0.034	-0.830	0.438
6	Nitrate	tm	-0.031	-0.950	0.337
7	OxyDis	tm	-0.032	-0.838	0.405
8	pH	tm	0.057	1.793	0.070
9	TDS	tm	-0.019	-0.504	0.617
10	Temperature	tm	-0.026	-0.740	0.472
11	Turbidity	tm	-0.010	-0.287	0.782
12	ImpactIndex	tm	-0.012	-0.312	0.767
13	Cond	tc1	0.020	0.671	0.495
14	COD	tc1	0.011	0.354	0.703
15	IronTot	tc1	-0.038	-1.174	0.255
16	PhosphoTot	tc1	0.000	0.021	0.988
17	Mangtot	tc1	-0.053	-1.286	0.201
18	Nitrate	tc1	-0.036	-1.147	0.264
19	OxyDis	tc1	-0.060	-1.666	0.104
20	pH	tc1	0.022	0.647	0.521
21	TDS	tc1	-0.010	-0.210	0.828
22	Temperature	tc1	0.018	0.541	0.594

23	Turbidity	tc1	-0.040	-1.079	0.26 5 0.43
24	ImpactIndex	tc1	-0.028	-0.844	6 0.51
25	Cond	tc2	0.019	0.652	5 0.78
26	COD	tc2	0.008	0.281	9 0.83
27	IronTot	tc2	-0.007	-0.183	9 0.63
28	PhosphoTot	tc2	-0.017	-0.473	9 0.21
29	Mangtot	tc2	-0.053	-1.266	5 0.32
30	Nitrate	tc2	-0.032	-1.007	6 0.23
31	OxyDis	tc2	-0.044	-1.192	1 0.64
32	pH	tc2	0.016	0.457	5 0.78
33	TDS	tc2	-0.013	-0.287	4 0.89
34	Temperature	tc2	-0.005	-0.127	7 0.77
35	Turbidity	tc2	-0.013	-0.333	6 0.95
36	ImpactIndex	tc2	-0.005	-0.061	6 0.86
37	Cond	op.gills	-0.009	-0.174	6 0.69
38	COD	op.gills	0.015	0.460	3 0.02
39	IronTot	op.gills	0.092	2.178	0 0.20
40	PhosphoTot	op.gills	-0.055	-1.275	7 0.88
41	Mangtot	op.gills	0.007	0.189	4 0.90
42	Nitrate	op.gills	0.003	0.127	0 0.01
43	OxyDis	op.gills	0.104	2.402	4 0.79
44	pH	op.gills	0.011	0.246	1 0.84
45	TDS	op.gills	0.006	0.195	6 0.23
46	Temperature	op.gills	-0.051	-1.161	8 0.00
47	Turbidity	op.gills	0.114	2.792	5 0.02
48	ImpactIndex	op.gills	0.097	2.485	0

49	Cond	Collector	0.023	0.670	0.537
50	COD	Collector	-0.032	-0.833	0.392
51	IronTot	Collector	0.022	0.636	0.547
52	PhosphoTot	Collector	-0.042	-1.196	0.226
53	Mangtot	Collector	0.039	0.913	0.404
54	Nitrate	Collector	-0.024	-0.705	0.494
55	OxyDis	Collector	0.044	1.255	0.209
56	pH	Collector	0.042	1.187	0.232
57	TDS	Collector	0.007	0.202	0.849
58	Temperature	Collector	-0.022	-0.620	0.530
59	Turbidity	Collector	0.061	1.692	0.102
60	ImpactIndex	Collector	0.038	1.057	0.282
61	Cond	Scraper	-0.033	-1.021	0.332
62	COD	Scraper	0.039	1.179	0.257
63	IronTot	Scraper	0.003	0.058	0.944
64	PhosphoTot	Scraper	0.057	1.744	0.076
65	Mangtot	Scraper	0.045	1.089	0.307
66	Nitrate	Scraper	0.021	0.662	0.518
67	OxyDis	Scraper	-0.008	-0.252	0.805
68	pH	Scraper	-0.001	-0.028	0.972
69	TDS	Scraper	-0.013	-0.395	0.735
70	Temperature	Scraper	0.010	0.314	0.773
71	Turbidity	Scraper	-0.061	-1.833	0.059
72	ImpactIndex	Scraper	-0.039	-1.232	0.259
73	Cond	Filterer	0.012	0.401	0.701
74	COD	Filterer	-0.011	-0.341	0.729

75	IronTot	Filterer	-0.023	-0.677	0.52 1
76	PhosphoTot	Filterer	-0.020	-0.592	0.58 5
77	Mangtot	Filterer	-0.082	-2.040	0.02 8
78	Nitrate	Filterer	0.001	-0.010	0.99 4
79	OxyDis	Filterer	-0.033	-0.947	0.35 3
80	pH	Filterer	-0.037	-1.152	0.25 6
81	TDS	Filterer	0.006	0.206	0.84 5
82	Temperature	Filterer	0.010	0.311	0.74 3
83	Turbidity	Filterer	0.006	0.131	0.89 6
84	ImpactIndex	Filterer	0.005	0.106	0.91 0